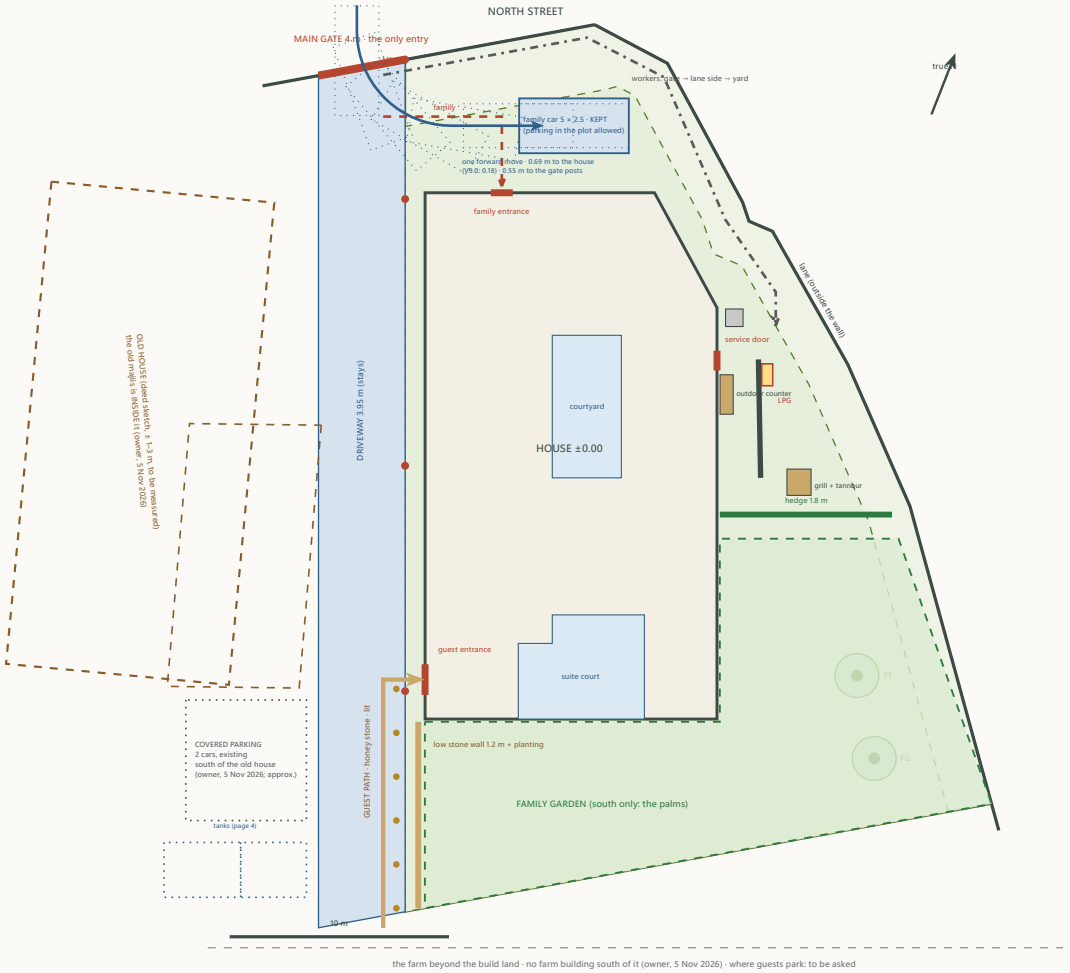
# Site and services plan · access, parking, paths, the family garden, the service yard, utilities, security, Civil Defense access, levels · Taif / Palm House V8.1 · STUDY

Design review S-01 … S-08, E-04 … E-06 · open item 20 · 2 Nov 2026 · drawn on `base.typ` (the plot, the M line, the gate, the palms); the old house (the old majlis inside it) and its annex from the deed sketch, its covered parking and the existing tank south of it west of the driveway near peg R, the septic outside the wall near C1 (the owner's answers, 5 Nov 2026) are drawn dotted, approximate, to be measured · **every site position is PROPOSED, to confirm with the official survey and the engineers**



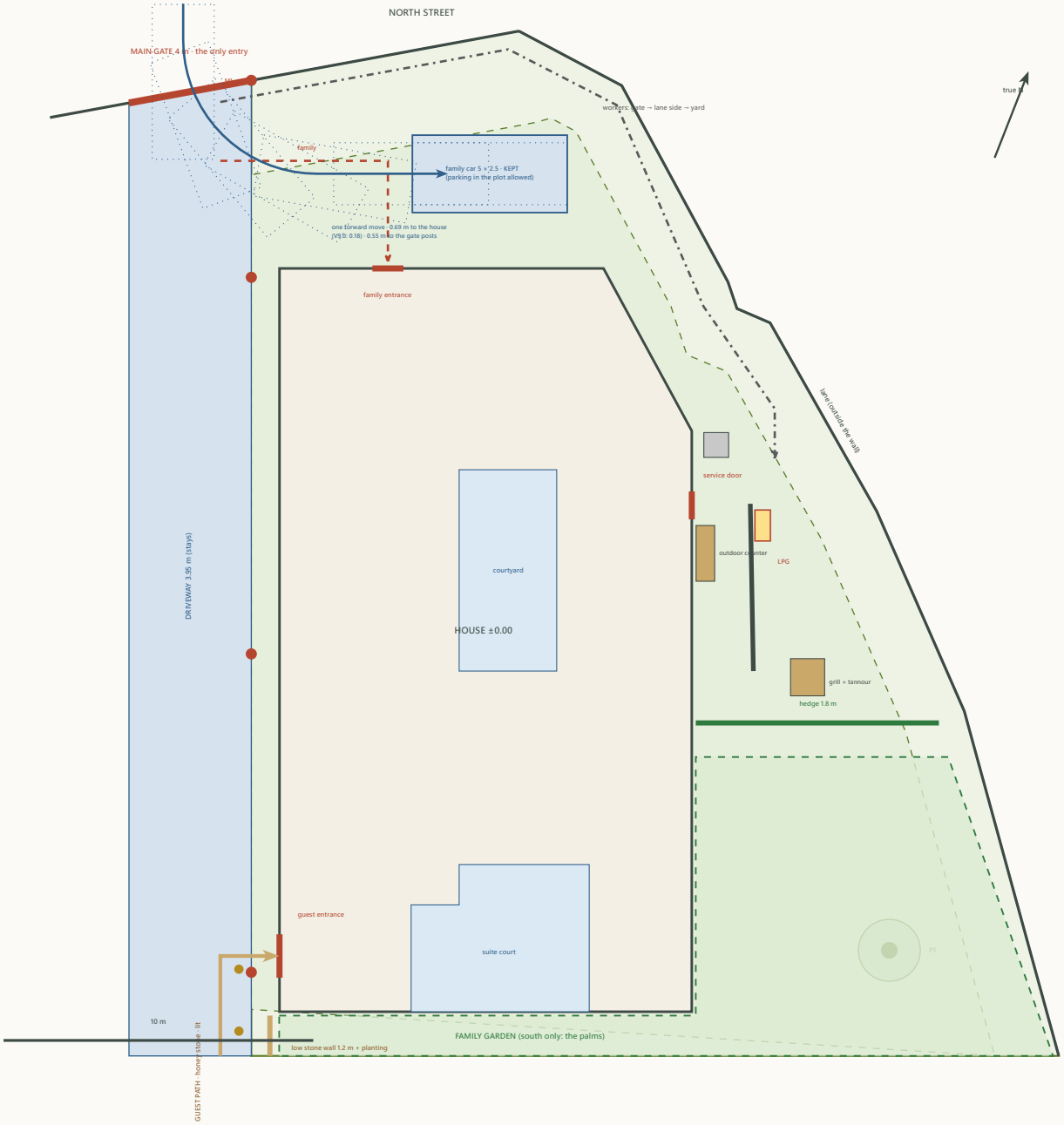

## 1 · Access and parking

- **One entry:** the farm is fully walled; the existing **4 m main gate** on the north street (peg M1 is its east post) is the only entry; no other gate (the lane-side gate of V8.0 is dropped). A **video intercom and a camera** at the gate (page 5).
- **The driveway** (3.95 m, west of the M line) stays: the cars' lane from the gate south to the family's existing **covered parking for two cars SOUTH of the old house, WEST of the driveway** (owner, 5 Nov 2026; drawn at x −10 … −4.5, y 29.2 … 34.7, approximate: its corners P1–P4 on the field sheet). **The car route:** gate → 29.2 m south on the straight driveway → a right turn west into the carport, which is open on its east side; the turn uses the driveway's full 3.95 m, so the guest path band is laid flush (no kerb) between y 27 and 37. Nothing of the new house touches this route. No new carport.
- **One uncovered family car KEPT** (owner, 2 Nov 2026: parking inside the plot is allowed by the municipality): a 5 × 2.5 bay in the north setback strip, 1.68 m off the house's north wall (8 Nov 2026: the bay on the car's turning line, y 1.77–4.27), 0.29 m east of the family entrance's path, which stays clear. **The swept path (8 Nov 2026, the owner's car test on site: a full forward turn into the north strip without any manoeuvre needs B1 6.00 m from M1):** the strip is now 6.1 m deep at the house's north-west corner (boundary wall to wall face; 3 m to the north setback line). The design car (assumed: a large family SUV 5.0 × 2.0 m, wheelbase 2.85, kerb-to-kerb turning circle Ø 11.8 m) comes in through the gate heading south on the line 2.2 m west of M1, starts turning left as it passes the gate and drives into the bay in **one forward move**: its body keeps **0.69 m from the house's walls** (V9.0: 0.18 m on the same line, too tight: the owner's result), 0.56 / 0.55 m from the gate's east (M1) and west posts and 0.84 m from the boundary wall; **the 4 m gate now governs, not the house**. The car leaves in reverse along the same line, or with a short reverse and a forward turn out through the gate. The manoeuvre crosses the family entrance path at grade (the kerb flush there), as on any forecourt.
- **Guests of the new house** park at the **driveway's south end** (5 Nov 2026: the owner's answers leave no farm building south of the build land; the brief's "farm parking" is not confirmed: **where guests park is to be asked**) and walk the **guest path**: a 1.2 m **honey-stone band along the driveway's east edge** from the driveway's south end north to the **guest entrance (the west foyer)**, 13.2 m, with bollard lights every 2 m; it never enters the family areas (the garden is behind its low wall, the north strip and the east yard are far). The driveway keeps 2.75 m beside it for one car. The band passes the tanks' fill point (y 37) and the carport's mouth (y 29.2–34.7) on the opposite (west) side of the driveway: flush there.
- **Guests of the old majlis** (owner, 5 Nov 2026: the old majlis is PART OF the old house, not a separate building): they go along the driveway to the old house, west of it, 3.8 m from the driveway's edge at the annex; they never reach the new house's paths or garden.
- **The family** enters by the north: gate → the north strip → the **family entrance** (8.8 m). **Workers and deliveries (3 Nov 2026, proposal):** gate → **along the farm's north boundary wall** (not along the house's north facade) → the lane-side strip (2.5–3 m wide east of the cut wall) → the gas cabinet and the bins at the **service yard** east of the screen (26.2 m), never through the garden, past the guest door, across the parking bay or through the outdoor kitchen: the north strip in front of the house stays a **calm family space** (the entrance path, the car).

## 2 · The family garden and its separation

- **A family garden on the south side only**, round the two palms, from the south facade to the plot's south line (≈ 236 m²); **no other garden** round the house: the north strip is the entrance and the optional car, the west strip the driveway, the east strip the service side.
- **Separation:** on the west a **low honey-stone wall 1.2 m with planting behind it** between the garden and the guest path on the driveway (a wall, not a hedge: it also holds the garden's soil 0.3 m above the drive and carries the path's lights); on the north-east a **1.8 m hedge** from the house's east wall to the boundary at y 20.7 between the service yard / workers' path and the garden (a hedge, not a wall: the yard's air and the lane-side breeze pass). The grill and tannour and the outdoor counter are inside the garden side.
- **No garden door from the living** (review S-04: **not needed, owner**): the **kitchen service door** is the family's door to the outside on the east; the garden is reached from it round the south-east corner (the hedge has a gate there) or from the guest entrance side.

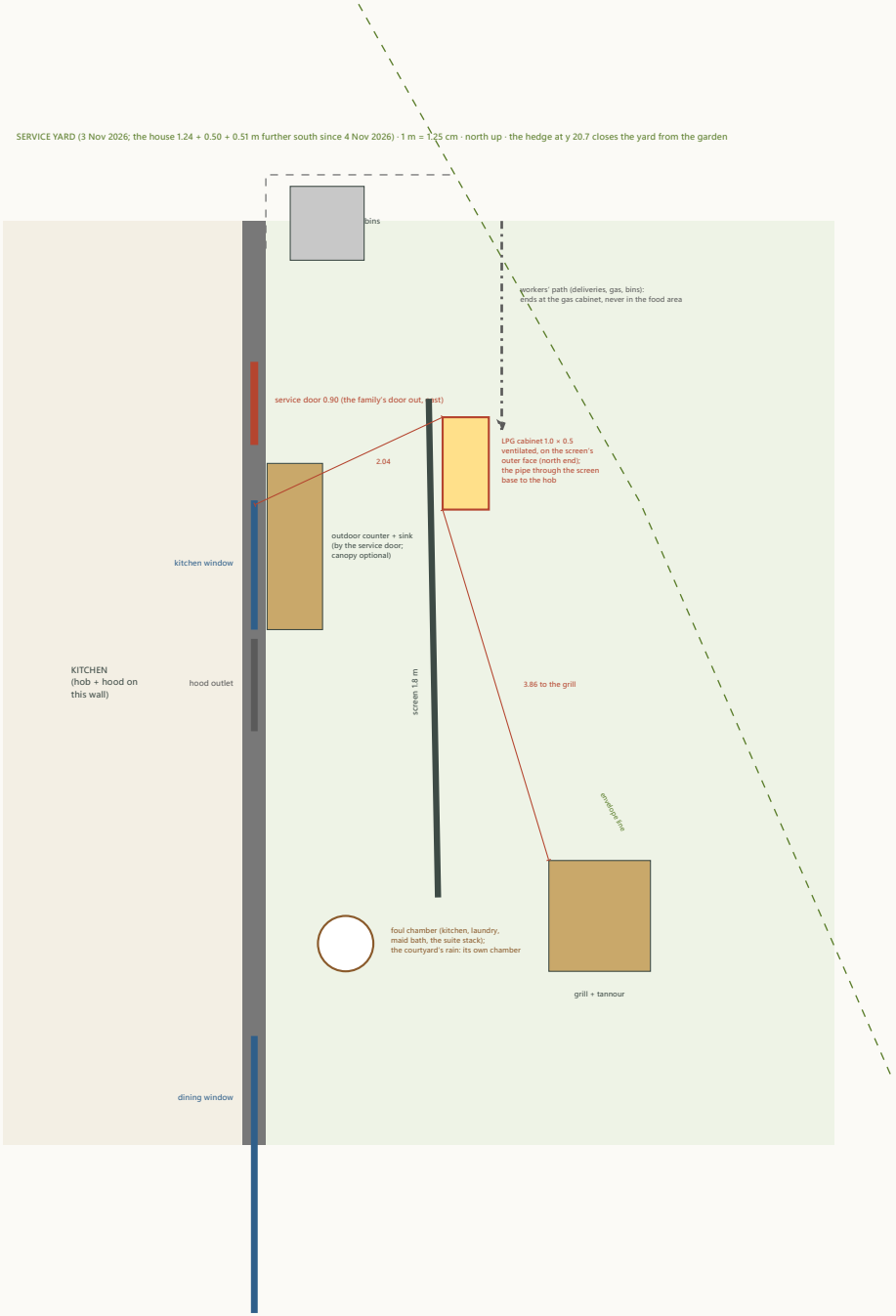
# Around the house (enlarged, y ≤ 31.5 m) · the strips, the paths, the family car (kept), the garden's edges




## Around the house

- **North strip** (6.1 m from the house's north wall face to the boundary wall at the north-west corner, 4 Nov 2026; 3 m to the north setback line): the family entrance path from the driveway, the family car (kept), the foul drain to the septic under its north edge (page 4); **3 Nov 2026:** the workers' path moved to the farm's boundary wall, so the strip in front of the house is the family's. **West strip** (0.785 m inside the M line) + the driveway: the guest path, the rescue ladder stand for bedrooms 2 and 3 (fire study P2: keep it open at night), the fire vehicle. **East strip** (2.5–9 m to the lane): the yard, the workers' path, the rain chamber (its soakaway in the garden's east corner since 8 Nov 2026). **South:** the garden.
- **Levels (page 4):** the house floor ±0.00 is assumed 0.30 m above the ground around it; every strip falls 2 % away from the walls; the driveway and the north strip fall to the gate; **to be set from the survey** (the street level at the gate, the farm's levels, the lane).

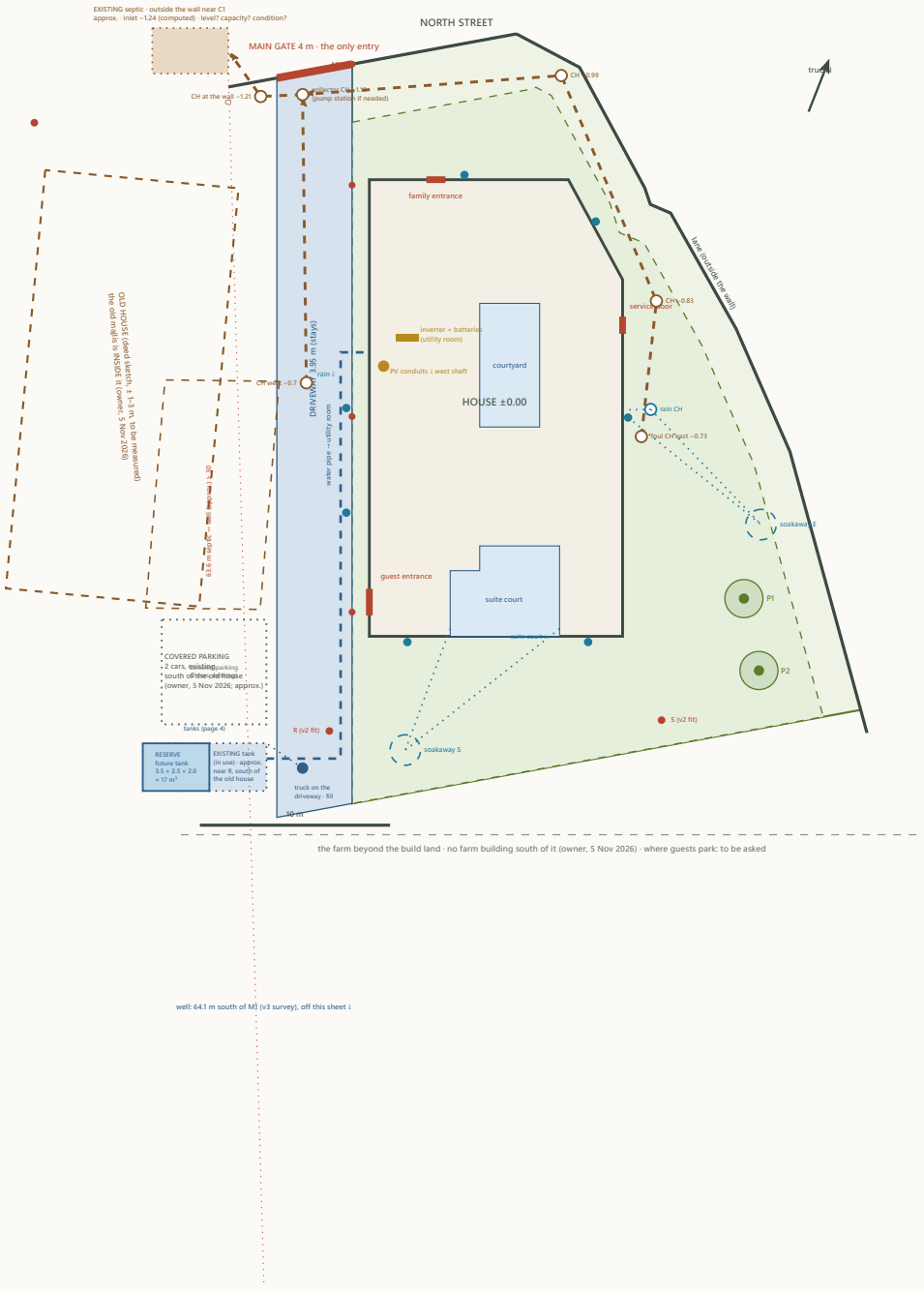
# The service yard (enlarged): the gas cabinet, the bins, the manhole, the screen, the workers' path · the rule used for the gas cylinders




## 3 · The service yard (east)

- **The outdoor kitchen (owner, 3 Nov 2026):** the **counter and sink NEXT TO the kitchen service door**, against the east wall south of it (CSV y 8.3–10.1, under the kitchen window; 0.24 m from the door), a **canopy over it optional** (shown dashed); the **grill and tannour a little apart** in the yard's south part (CSV x 16.5–17.6, y 12.6–13.8): 5.52 m from the service door, 4.05 m from the kitchen window, 3.48 m from the hood outlet, 3.26 m from the dining window, 12.6 m from the HRV intake, downwind of the house in the north-west wind, 8.5 m from the nearest palm. **The yard re-planned:** the **1.8 m screen** moved east to CSV x 15.2 (its north end at y 7.6 on the envelope line) so the counter keeps a 1.15 m working strip; the **bins** north of the service door (CSV y 5.3–6.1), outside the food area; the **gas cabinet** on the screen's outer face at its north end, 3.86 m from the grill; the **manhole** (the foul chamber) at the yard's south end; the workers reach the gas and the bins from the lane-side strip without crossing the outdoor kitchen.
- **The gas cylinders:** in a **ventilated outdoor cabinet 1.0 × 0.5 on the screen's outer (east) face at its north end** (CSV x 15.35–15.85, y 7.8–8.8), so the delivery man changes cylinders from the workers' path without entering the yard; the pipe through the screen's base to the hob on the east wall (≈ 3 m). **The rule used (assumed, to confirm with the Civil Defense):** NFPA 58 as adopted in Saudi practice for small cylinders outdoors: ≥ **1.5 m** from any opening into the building below the cylinder's relief valve (doors, windows, the hood outlet), ≥ **3 m** from ignition sources, mechanical air intakes, drains and manholes (LPG is heavier than air), the cabinet ventilated top and bottom, the cylinders upright, the regulator and the shut-off outside, the gas detector and the automatic valve inside (fire study). **Measured:** the service door 2.04 m, the kitchen window 2.04 m, the hood outlet 2.47 m, the dining window 6.05 m, the manhole 4.82 m, the grill 3.9 m, the HRV intake on the roof > 10 m, the lane boundary 3 m. **OK**
- **Screened from the guests' path:** the yard is on the east, the guest path on the west; nothing of the yard is seen from it. The **workers' path** (3 Nov 2026 proposal: along the farm's boundary wall, then the lane-side strip) ends at the gas cabinet and the bins east of the screen, never in the outdoor kitchen; the **hedge** at y 20.7 stops it short of the garden.

# Utilities · reserve positions and routes, no system design · water, sewage, power (PV only, no generator), rainwater · levels



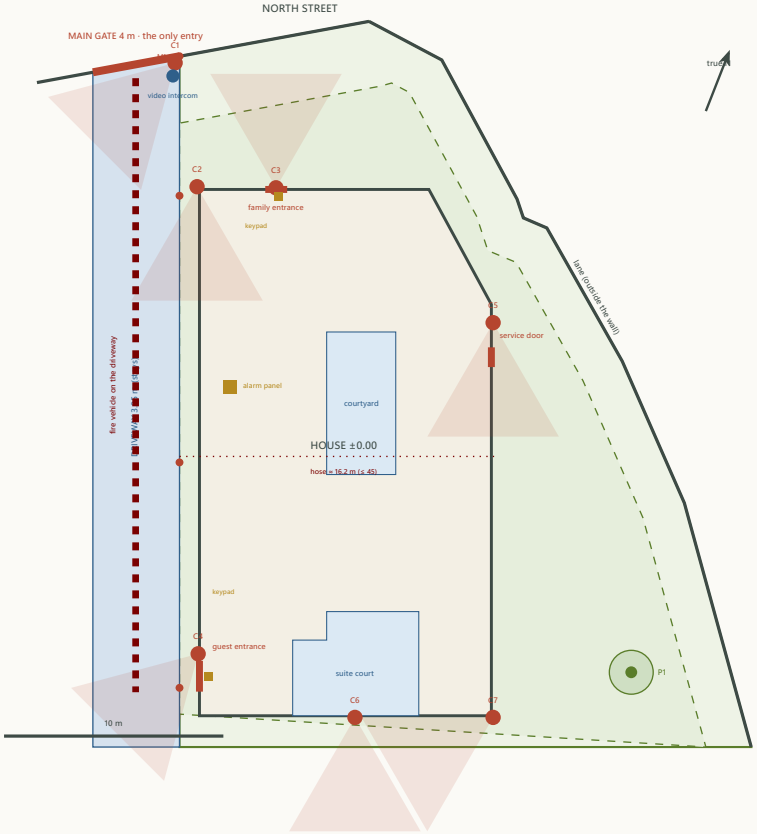

## 4 · Utilities

- **Water (owner, 5 Nov 2026):** the **existing ground tank (in use) is SOUTH of the old house, WEST of the driveway, near peg R** (not in the new house's garden): drawn at x −7.5 … −4.5, y 35.7 … 38.2, **approximate, to be measured** (its lid and fill point on the field sheet), 3.4 m from R and 1 m south of the covered parking (no clash **OK**). **The reserved future tank** (3.5 × 2.5 × 2.0 m, ≈ 17 m³) goes **NEXT TO it, west of it**, also west of the driveway. The **water truck** stands on the driveway beside the tanks (y 33–39; the guest path band flush there) and fills through the fill point at the existing tank's north-east corner, 2.3 m of hose. **The supply pipe to the utility room** (the manifold, the heater, the main valve) leaves the existing tank eastwards, **crosses the driveway at y 36.5 in a sleeve under the drive** (4 m), then runs north under the guest path band on the driveway's east edge and through the west wall: 26.8 m assumed (the engineer checks the pump and the pipe size; the new tank shares the pipe). The tanks' nearest corner is 7.8 m from the new house's south-west corner and its footings (≥ 5 m **OK**), 7.3 m from the south soakaway (rain): the **south garden** is free of them (the brief's tank "east of peg R" is withdrawn). The north strip stays free (the turning pocket for the car, page 1). Also the fire reserve, if the Civil Defense asks for one.
- **Sewage (owner, 3 and 5 Nov 2026):** the **existing septic tank is OUTSIDE the farm wall near corner C1**, beside the old house, and **SERVES THE OLD HOUSE**: kept there, **approximate, to be measured on site** (its lids, its inlet(s), the old house's connection). The new house's foul drains of both clusters run to it NORTH and WEST, **crossing the driveway once** (the north strip's chamber → the collector on the driveway, under the family entrance's path) with **a chamber at every change of direction**: the **east main** from the foul chamber in the yard (the kitchen, the laundry, the maid bath, the suite stack) north under the workers' path round the cut wall (chamber at the bend), along the north strip's north edge (chamber), to the **collector chamber on the driveway inside the gate**; the **west branch** from the west chamber on the driveway (the west wash block's run, which now falls **north** inside the house) to the same collector; then west on the driveway to the **chamber at the wall** (inside the wall, 0.9 m from it) and **under the boundary wall 1.4 m west of the gate's post** to the septic's inlet: the **east main 38.8 m** from the yard's chamber to the inlet (33.7) m to the collector + 5.1 m out), the **west branch 20.3 m** from the west chamber. **Falls, all ASSUMED (the plumbing engineer):** Ø 110 branches 1.5 %, the Ø 160 main 1.0 %, 0.03 m through each chamber, the invert at the farthest fixture −0.50 (the crown 0.10 under the ground slab): foul chamber east −0.73 → bend −0.83 → north strip −0.99 → collector −1.16 (the west branch arrives at −0.88) → the wall chamber −1.21 → **the septic's inlet at −1.24 below the house floor**, ≈ 0.84 m below the assumed ground at the gate. **So the septic's inlet level must be surveyed:** if it lies higher than ≈ −1.24 (the house floor ±0.00 is itself to be set against the street), the main cannot fall to it and a **small packaged pump station** (two pumps, an alarm) at the collector chamber would be needed: flagged. **Flags for the survey, the municipality and the engineer:** the septic's **legal position outside the wall** (under the street side: the municipality's consent to keep using it and to cross under the wall), its inlet level, its **capacity for BOTH houses** (the old house's load that it already takes + the new house's ≈ 12 persons with staff: the engineer sizes it or adds a second tank / a treatment unit), its condition and age, **whether a pump is needed** (the inlet level against −1.24); its vent (page 6 of the plumbing study: away from the house, the gate and the north parking).
- **Power: solar PV provision only** (the roof zones reserved: Z4a / Z4b on the south wing, Z5 on the north block); the **conduits** from the roof come down the **west shaft** to the **utility room**, where the **inverter and the battery cabinet** go (owner, 2 Nov 2026: the utility room; the electrical engineer confirms the space, the ventilation and the fire safety; the room's layout with the water kept away from the batteries is in `studies/plumbing.typ`). **No generator** (review E-05: **no generator, owner**). **Note:** until the PV and the batteries are installed there is **no backup during power cuts** (the lift, the HRV, the pumps and the gas valve stop; the gas valve fails closed).
- **Rainwater — fully separate from the sewage (3 Nov 2026, corrected):** the roof's 11 outlets reach the ground in internal downpipes and discharge at the plinth: north (1), west (2, into the driveway's channel), the cut and the east wall (2, into the **east rain chamber** and the **east soakaway**, moved on 8 Nov 2026 out of the narrow east strip into the garden's east corner, 6.4 m from the walls), south (2, into the garden's soakaway); the **courtyard's drain and overflow** leave the east wall to the **east rain chamber** (their own chamber 1.5 m away, not the foul chamber: corrected); the **suite court's gutter outlets** come down the gym's and the office's walls to the **south soakaway** in the garden; the site's surface water falls to the soakaways and the driveway's channel to the gate. **Nothing of the rain ever enters the septic.** The **final discharge** is chosen with the soil report (open item 7), the levels, the well (soakaways ≥ 10 m from it, assumed rule) and the foundations (≥ 5 m from the house, assumed rule): the two soakaways as drawn, each with a **safe overflow** at the surface towards the lane side / the driveway's channel, never towards the house.

## 7 · Levels (to be set from the survey)

- **Assumed:** the house floor ±0.00; the ground around the house at −0.30 (the plinth 0.6 m shows 0.3 above it); the courtyard's seating −0.15 inside; every strip falls 2 % away from the walls for 2 m, then towards the soakaways; the driveway and the north strip fall gently to the gate (the street level unknown); the garden at −0.30 … −0.60 with the palms' existing ground kept; the yard at −0.30 with its manhole; the old house, its carport and the tanks: unknown.
- **The ground falls slightly to the south (owner, 8 Nov 2026); the owner accepts raising the house on a plinth if needed.** The finished floor level, the plinth's height, the steps or a ramp at the family and guest entrances, the courtyard's and the garden's levels and the sewer falls are **to be set from the official survey's levels** (`docs/SITE_SETOUT.md` § 9, `docs/OPEN_ITEMS.md`).
- **From the survey:** the street level at the gate, the plot's levels at the pegs M1–M4, corner C1 and peg R, the lane's level along the east boundary, the palms' trunk bases, the old house's floor, the septic's position, lids and **inlet level** (outside the wall near C1; the old house's connection), the tanks' positions, lids and fill point, the carport's corners, the well's head. Then the engineer sets the finished floor against the street and the drainage falls.

# Security · Civil Defense access · the items for the survey, the municipality and the engineers




## 5 · Security

- **The main gate:** a video intercom (the only entry) and camera C1 on the gate post looking at the gate and the street; the gate's motor and the intercom on the house's alarm circuit.
- **Cameras round the house** (black fittings, high on the walls under the band or the cornice, IR at night): **C1** the gate (with the video intercom); **C2** the driveway, looking south; **C3** the family entrance; **C4** the guest entrance and the path; **C5** the service yard, looking south; **C6** the garden and the south facade; **C7** the suite court's outside (the screen) and the east garden. Together they cover the gate, the driveway, the family entrance, the guest entrance, the service yard, the garden and the suite court's outside (its mashrabiya screen and the south-east garden). The recorder with the **alarm panel in the utility room**; keypads at the family entrance and the guest entrance; contacts on the three exit doors and the suite court's doors; the gas detector and the smoke alarms on the same panel (fire study). **The electrical engineer** designs it; positions proposed.

## 6 · Civil Defense access

- **Can a fire vehicle reach the house?** The gate is 4.02 m (≥ 3.5 **OK**), the driveway 3.95 m wide and straight for 39 m (≥ 3.5 **OK**): a vehicle stands on the driveway along the whole west facade, 0.8 m from it, and sees the north facade from the street and the gate. **There is no turning area inside the plot**: the vehicle turns at the driveway's south end in the farm beyond (to confirm that the ground there is open and firm; 5 Nov 2026: no farm building south of the build land) or reverses out; the carport's mouth west of the driveway (y 29–35) gives it a swing. **Hose reach:** from the driveway round the house to the east wall ≈ 16.2 m, the farthest point of the house (≤ 45 m **OK**). The east facade is also reached from the **lane** outside the farm wall (2.5–9 m away) if the lane is drivable (to confirm), the south facade from the driveway's south end.
- **The engineering office must confirm** (SBC 201 / the Civil Defense's villa checklist): the required approach distance to the facades, the hose reach, the water supply (the tank as a fire reserve, a hydrant on the street?), the gate's clear height, and the rescue ladder stand on the driveway for bedrooms 2 and 3 (fire study P2). **to confirm**

## Items for the survey

1. The old house (the old majlis inside it) and its annex, the covered parking south of it, the existing ground tank near peg R (lid, fill point, pump), the well, the septic tank (outside the wall near C1: its lids, inlet level, size, the old house's connection) and the farm's walls: their positions and levels (all approximate here; the field sheet in `docs/SITE_SETOUT.md` § 8). Where guests park.
2. The street level at the gate, the levels at the pegs, the lane's level, the plot's south line, the palms' trunks.
3. The legal boundaries and the setback lines (open item 1): the north street line and the lane line against the envelope used here.

## Items for the municipality

1. Parking in the north setback strip (one uncovered car): allowed?
2. The septic tank's reuse for the new house and its **legal position outside the wall** (under the street side near C1), the drain crossing under the wall; the rain soakaways and their overflows; the discharge to the lane side.
3. The gas cabinet and the Civil Defense's villa requirements (access, water, alarms).

## Items for the engineers

1. Civil / MEP: the septic's capacity for BOTH houses (the old house's load and the new house's ≈ 12 persons), its condition and inlet level (a pump station at the collector if it is too high), the sewer falls as computed here, the separations (the rules assumed here), the well distance; the water tank's pump and pipe; the new tank's size and the fill point; the soakaways from the soil report; the PV conduits and the inverter / battery space; the gas pipe and the cabinet's details; the levels and the drainage round the house.
2. Electrical: the cameras, the intercom, the alarm panel, the gate motor, the night lights on bedroom 1's route, the outdoor lighting of the paths.

**Study checks passed** (19)

Checks: the design car (5.0 × 2.0, Ø 11.8 m) turns from the gate into the north strip in ONE forward move (owner's car test, 8 Nov 2026): 0.69 m to the house's walls (V9.0: 0.18), 0.56 / 0.55 m to the gate posts, 0.84 m to the boundary wall (every ≥ 0.5) · the soakaways ≥ 5 m (edge) from the house's walls (east 6.35, south 5.05), ≥ 1 m inside the deed line (1.68 / 1.46) and ≥ 10 m from the well (assumed rules, the soil report decides) · the gas cabinet ≥ 1.5 m from every opening of the house (service door 2.04, kitchen window 2.04, hood outlet 2.47, dining window 6.05) · the gas cabinet ≥ 3 m from the manhole (4.82) and the grill (3.9) · the gas cabinet ≥ 1.5 m from the lane boundary (3) · the family parking bay (kept) inside the plot, 1.68 m off the house's north wall (≥ 0.4), 0.29 m clear of the family entrance's path · the guest path never enters the family garden (west of its low wall) and ends at the guest door · the workers' path stays north of the garden's hedge and east of the house · the tanks (existing + reserve) ≥ 5 m from the house's south-west corner and its footings (7.8 m, assumed positions) · the septic OUTSIDE the plot's wall near C1 (every corner outside the land polygon), its inlet ≥ 1.0 m below the house floor on the computed falls (−1.24) and −0.84 m below the assumed ground at the gate · the septic ≥ 15 m from both water tanks (35.2) and the foul chambers ≥ 15 m from them (19.1) · the water pipe ≥ 1.5 m from the foul drains along the driveway (1.8 m) and from the rain chambers · the rain system never joins the foul one: the courtyard's outlet reaches the rain chamber (1.5 m), not the foul chamber · the gate (4.02 m) and the driveway (3.95 m) take a fire vehicle (≥ 3.5 m); every point of the house within 45 m of hose from the driveway (≈ 16.2 m) · the outdoor counter next to the service door (0.24 m, ≤ 2 m) with a 1.0 m working strip to the screen (1.15 m); the grill ≥ 3 m from the service door (5.52), the kitchen window (4.05), the hood outlet (3.48), the dining window (3.26), the maid window (8.97), the gas cabinet (3.86) and ≥ 10 m from the HRV intake (12.6) · the workers' path (along the farm's boundary wall, 3 Nov 2026 proposal) never crosses the outdoor kitchen (the counter + its working strip, the grill) nor the parking bay nor the family entrance path · the grill ≥ 3 m from the palms (8.5)) and the gas cabinet (3.9) · the septic ≥ 30 m from the well (63.6 m): both positions approximate, from the survey · the covered parking and the tanks (owner, 5 Nov 2026) lie WEST of the driveway and SOUTH of the old house (its south edge y 28.7, deed sketch), 1 m apart (no clash), the existing tank 3.4 m from peg R; the truck stands on the driveway; the pipe crosses the driveway once. Every position is proposed, to confirm with the official survey and the engineers; the existing buildings are approximate (the old house from the deed sketch; its carport and the tanks south of it west of the driveway near peg R; the septic near C1: the owner's answers of 5 Nov 2026, to be measured on site).